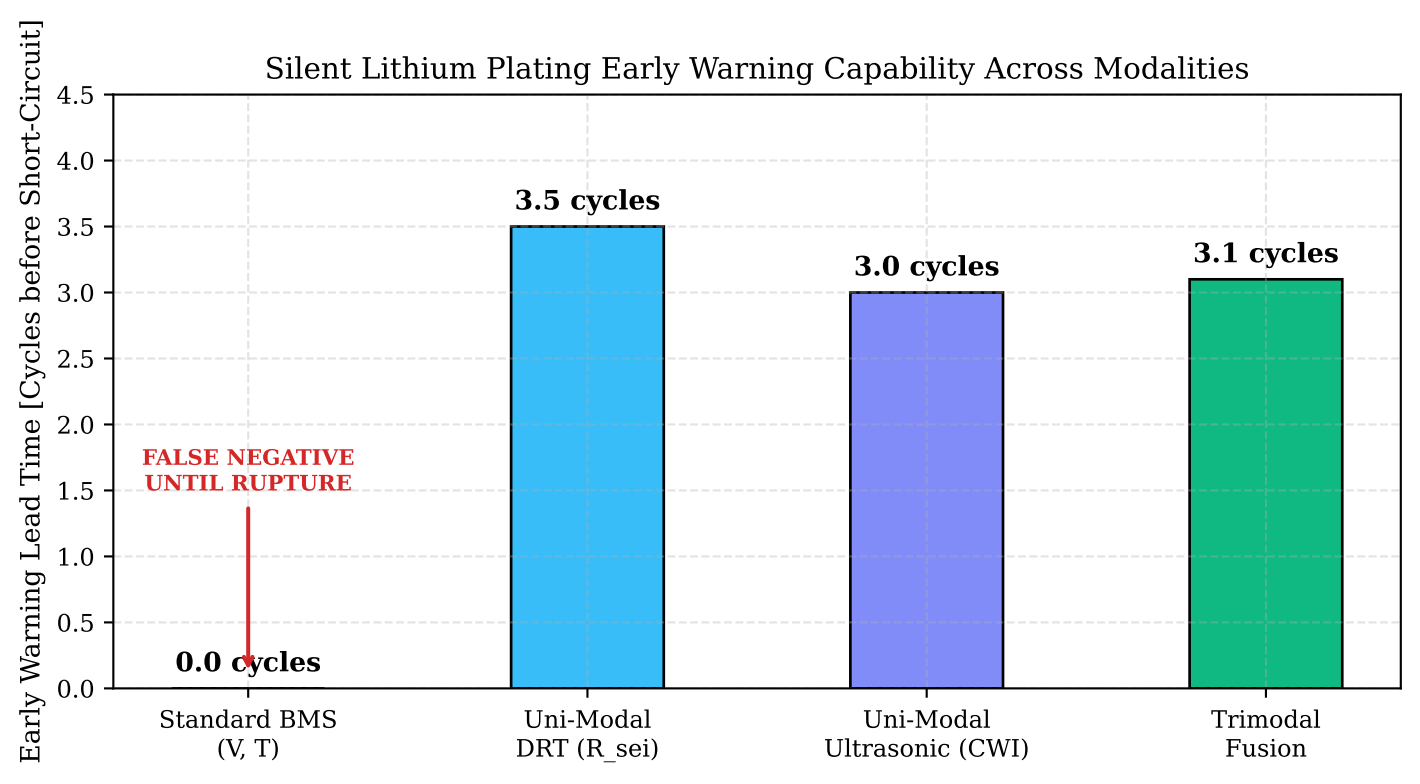
Silent Lithium Plating Early Warning Capability Across Modalities
Early Warning Lead Time [Cycles before Short-Circuit]
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
3.5 cycles
3.0 cycles
3.1 cycles
FALSE NEGATIVE
UNTIL RUPTURE
0.0 cycles
Standard BMS
(V, T)
Uni-Modal
DRT (R_sei)
Uni-Modal
Ultrasonic (CWI)
Trimodal
Fusion